Yes Bank Technical Documentation

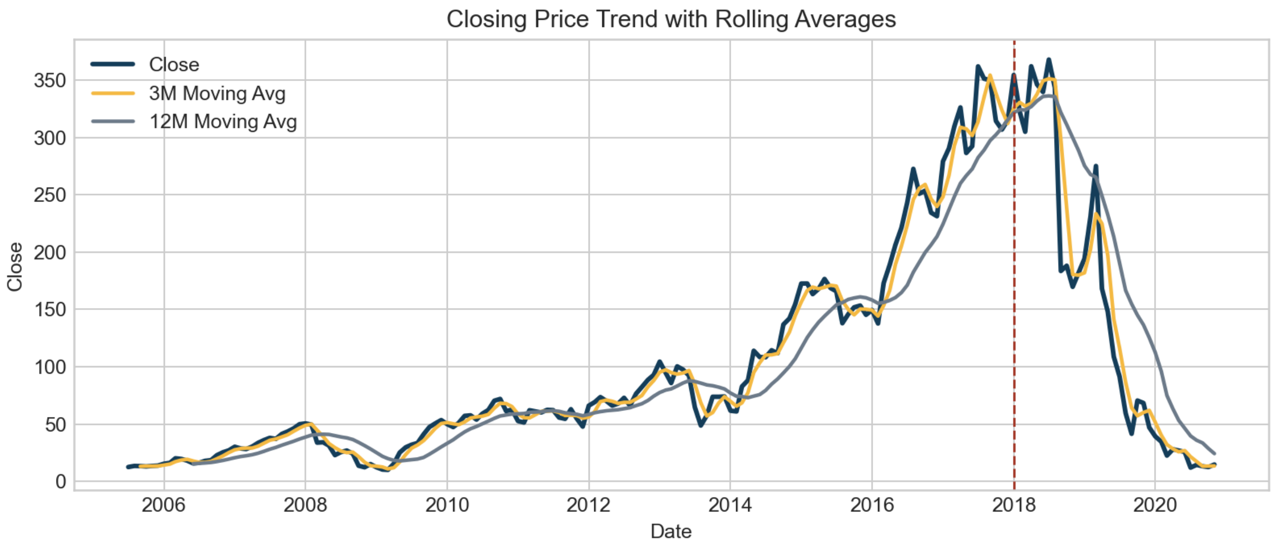
This document describes the standalone Yes Bank stock closing price prediction project, including business framing, EDA, preprocessing, model comparison, evaluation summary, and final conclusions.

1. Abstract

The project predicts monthly closing price using OHLC data and engineered time-aware features. The same dataset used by the reference project is retained, so the main EDA conclusions remain aligned: high correlation among price fields, visible positive skew, and strong multicollinearity. The updated workflow adds deeper regime analysis and a more explicit holdout evaluation story.

2. Data Description

Attribute	Value
Rows	185
Columns	10
Start Month	Jul 2005
End Month	Nov 2020
Missing Values	13
Duplicate Rows	0

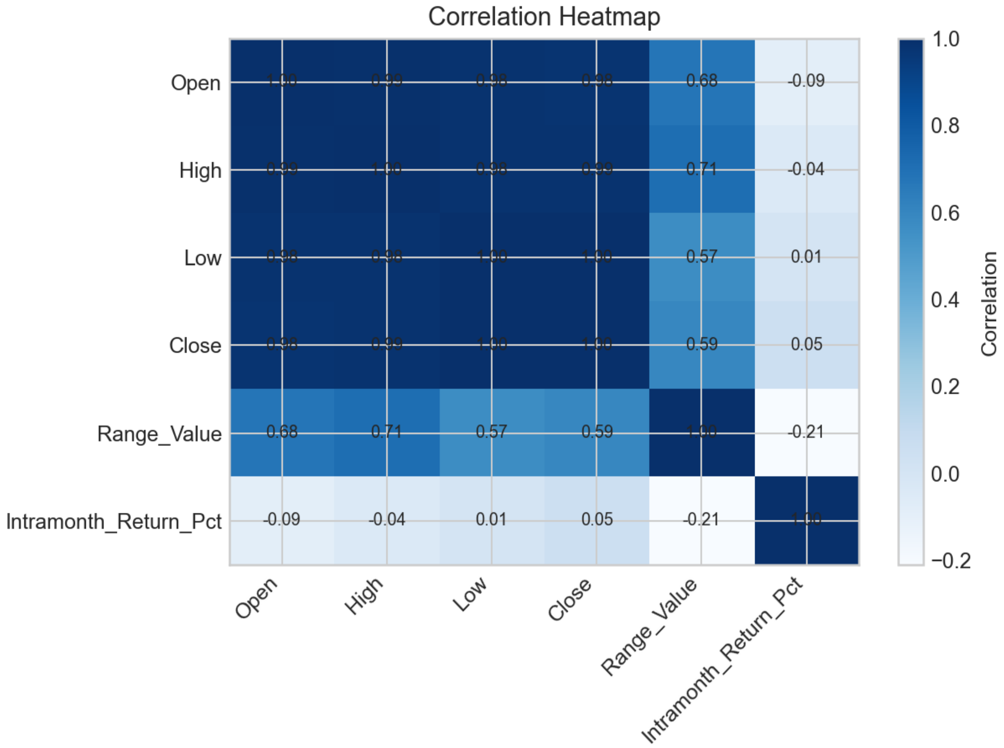


3. Exploratory Data Analysis

- All four price variables are positively skewed, which is consistent with the stock’s long rise followed by a sharp reset.
- The strongest raw correlation with close comes from Low, followed by High and Open.
- The most volatile months cluster around the 2018 crisis and the 2020 COVID shock.
- The average monthly range is much larger after 2018, confirming a structural regime change.

Feature	Correlation with Close
Low	0.9954
High	0.9851
Open	0.9780
Range_Value	0.5947

Feature	Correlation with Close
Intramonth_Return_Pct	0.0505



4. Feature Engineering and Preprocessing

Feature creation includes monthly range statistics, open-high-low gaps, lagged close values, rolling mean and rolling standard deviation features, cyclical month encoding, and regime flags for post-2018 stress and the COVID period. Instead of dropping core price variables to fight multicollinearity, the workflow keeps them and lets regularization absorb the instability.

Feature 1	Feature 2	Correlation
Open	High	0.9930
Open	Low	0.9840
High	Low	0.9830
Open	close_lag_1	1.0000
High	close_lag_1	0.9930
Low	close_lag_1	0.9840
Open	close_lag_3	0.9530
High	close_lag_3	0.9610
close_lag_1	close_lag_3	0.9530
Open	close_ma_3	0.9870
High	close_ma_3	0.9870
Low	close_ma_3	0.9680

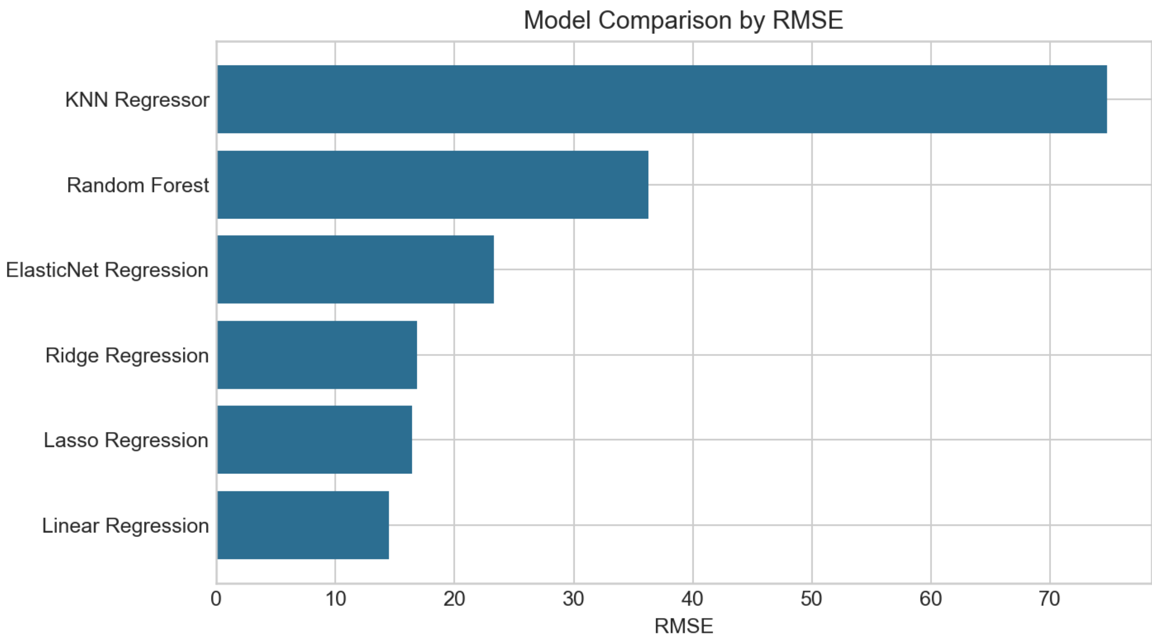
Target Strategy	MAE	RMSE	R2
Raw Close	11.05	16.43	0.9834
log1p(Close)	71.84	82.87	0.5771

Target conditioning checks show that the raw close performs better than log-transformed close on the chronological holdout set, so the final notebook keeps the target on its natural rupee scale.

5. Model Implementation

The core notebook comparison uses a chronological 80:20 split and TimeSeriesSplit-based hyperparameter tuning. This keeps the evaluation closer to a realistic time-aware setup while preserving the same overall problem framing as the reference project.

Model	MAE	RMSE	R2
Linear Regression	10.08	14.48	0.9871
Lasso Regression	11.05	16.43	0.9834
Ridge Regression	11.48	16.83	0.9826
ElasticNet Regression	15.78	23.30	0.9666
Random Forest	28.36	36.28	0.9189
KNN Regressor	58.70	74.81	0.6554



Best tuned parameters for the regularized family: Lasso {'model__alpha': 0.2}, Ridge {'model__alpha': 1.0}, ElasticNet {'model__alpha': 0.2, 'model__l1_ratio': 0.9}.

6. Evaluation Summary

The main outcome stays aligned with the reference project in an important way: the linear and regularized linear family is the strongest part of the benchmark. In this standalone version, Lasso is selected for deployment because it keeps errors low and remains easy to explain in an interview setting.

Date	Actual	Predicted	Residual	Absolute Error
Dec-2017	315.05	311.14	3.9085	3.9085
Jan-2018	354.45	351.98	2.4688	2.4688
Feb-2018	322.25	325.12	-2.8684	2.8684
Mar-2018	304.90	297.44	7.4614	7.4614
Apr-2018	362.05	357.05	4.9998	4.9998
May-2018	346.20	334.51	11.69	11.69

Date	Actual	Predicted	Residual	Absolute Error
Jun-2018	339.60	332.73	6.8659	6.8659
Jul-2018	367.90	377.39	-9.4870	9.4870
Aug-2018	343.40	371.67	-28.27	28.27
Sep-2018	183.45	207.35	-23.90	23.90
Oct-2018	188.05	244.47	-56.42	56.42
Nov-2018	169.65	189.58	-19.93	19.93

7. Model Explainability

Feature	Coefficient	Absolute Coefficient
Low	61.74	61.74
High	19.59	19.59
high_open_gap	6.3069	6.3069
year	0.8841	0.8841
month_sin	0.5332	0.5332
range_pct	-0.2015	0.2015
close_std_6	-0.0862	0.0862
Open	0.0000	0.0000
close_std_3	-0.0000	0.0000
post_2018_crisis	0.0000	0.0000
time_index	0.0000	0.0000
month_cos	-0.0000	0.0000

The most influential features are still dominated by same-month price levels and short-memory close history. That is consistent with the central intuition of the dataset: the monthly close is tightly anchored to contemporaneous price structure.

8. Additional Findings

Date	Range_Value	Close	Intramonth_Return_Pct
Sep-2018	183.85	183.45	-47.16
Apr-2019	120.60	168.00	-39.35
Oct-2018	99.85	188.05	4.4700
Nov-2018	84.40	169.65	-11.64
Mar-2020	82.40	22.45	-36.22

- The strongest downside shocks line up with known stress windows, not random isolated months.
- The data is simple in shape but not in regime behavior; that is why validation strategy matters more than brute-force model complexity.
- The updated Streamlit dashboard adds a more presentation-ready layer than the reference repo by surfacing trend, volatility, correlation, and prediction quality in one place.

9. Final Conclusion

Because the dataset is the same, the broad analytical outcome remains intentionally close to the reference project: the data is clean, price variables are highly correlated, regularized regression is a strong fit, and the 2018 crisis materially changes the stock behavior. The main difference in this standalone repository is that the analysis is more explicit about regime change, chronological validation, and model explainability. That makes the project stronger for interviews while still remaining faithful to the underlying dataset story.